Hypertension

ORIGINAL ARTICLE

Cluster Randomized Controlled Trial of Intensive Systolic Blood Pressure Control in Patients With Renal Cell or Thyroid Cancer Receiving VEGFR Tyrosine Kinase Inhibitors: ECOG-ACRIN EAQ191

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BACKGROUND: The objective of this cluster randomized controlled trial was to determine the feasibility and safety of an intensive (systolic blood pressure [SBP] <120 mm Hg) versus usual care (SBP <140 mm Hg) approach to SBP control in patients with renal cell or thyroid cancer initiating VEGFR (vascular endothelial growth factor)-tyrosine kinase inhibitors.

METHODS: A phase II site–based cluster randomized controlled trial compared intensive SBP control to usual care, incorporating a centralized BP advisory core to guide BP management. Patients underwent home BP monitoring and study visits at baseline, 1, 2, 3, and 6 months. SBP was compared between the 2 study arms using bootstrapped CIs. Common Terminology Criteria for Adverse Events and patient-reported outcomes were summarized.

RESULTS: Overall, 61 patients with renal cell (n=58) or thyroid cancer (n=3) from 10 sites were enrolled; 30 at 5 sites were randomized to intensive SBP control and 31 at 5 sites to usual care. A lower SBP was observed in the intensive SBP control arm compared with usual care, with mean differences of −12.2 (95% CI, −18.1 to −7.0) mm Hg at 1 month, −7.6 (95% CI, −15.3 to −0.4) mm Hg at 3 months, and −6.9 (95% CI, −19.3 to 6.0) mm Hg at 6 months. In the usual care arm, Grade 3 Common Terminology Criteria for Adverse Events for kidney injury (n=4), hypotension (n=3), and dyspnea (n=2) were numerically greater compared with the intensive SBP control (n=1, 2, and 0, respectively). Patient-reported outcomes were largely similar between the 2 groups.

CONCLUSIONS: This first-ever randomized controlled trial of SBP control in an active cancer population demonstrates the feasibility, safety, and tolerability of intensive SBP control with VEGFR tyrosine kinase inhibitors.

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Key Words: blood pressure ■ carcinoma, renal cell ■ cardio-oncology ■ cardiotoxicity ■ hypertension ■ neoplasms ■ tyrosine kinase inhibitors

Over the past decade, VEGFR (vascular endothelial growth factor) tyrosine kinase inhibitors (TKIs) have dramatically improved the survival rates of patients with certain cancers, including renal cell cancer (RCC) and thyroid cancers. VEGFR TKIs represent a standard first-line approach in metastatic RCC, most commonly in combination with other potentially cardiotoxic therapies, such as immune checkpoint inhibitors.[1] Similarly, VEGFR

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ORIGINAL ARTICLE

NOVELTY AND RELEVANCE

What Is New?

This is the first-ever study in an active cancer population with hypertension receiving VEGFR (vascular endothelial growth factor) tyrosine kinase inhibitors that reached its primary end point of a significant difference in systolic blood pressure (BP) with intensive systolic BP control compared with usual care. Our site-based cluster randomized trial was novel through (1) its use of a central BP advisory core; (2) home BP monitoring; (3) integration of a platform that was able to receive home BPs, allow for biweekly patient notifications and reminders to increase compliance of BP measurements, and issue alerts for patients and study staff of any BP alarm thresholds; and (4) geographic and institutional diversity, recruiting patients across the country from both university- and community-based health systems to emphasize relevance with trends in national cancer care delivery.

What Is Relevant?

Using a site-based cluster randomized trial design, our study demonstrates that intensive systolic BP control is feasible, safe, and well-tolerated with fewer adverse events than usual care in an advanced cancer population with cancer treatment–associated hypertension.

Clinical/Pathophysiological Implications?

Our findings lay the necessary foundation to test the implementation of an intensive systolic BP control strategy in larger active cancer populations.

Nonstandard Abbreviations and Acronyms

BP	blood pressure
CTCAE	Common Terminology Criteria for Adverse Events
CVD	cardiovascular disease
HF	heart failure
ICC	intraclass correlation coefficient
MACE	major adverse cardiovascular event
NCORP	National Community Oncology Research Program
PRO	patient-reported outcome
RCC	renal cell cancer
RCT	randomized controlled trial
SBP	systolic blood pressure
TKI	tyrosine kinase inhibitor
VEGFR	vascular endothelial growth factor receptor

TKIs have provided much needed therapeutic options for patients with radioactive iodine-refractory differentiated or medullary thyroid cancers.[2] These therapies are used indefinitely for the long-term control of RCC and thyroid cancer, with current overall median survival rates on the order of 46 to 56[3] and 26 to 41 months,[4] respectively. Thus, maximizing their tolerability to enable durable and consistent treatment adherence is increasingly important.

VEGFR TKIs are associated with important cardiovascular risks.[5] Hypertension secondary to VEGFR TKIs is common and affects a significant proportion of patients.[6] In phase III RCC randomized controlled trials (RCTs), nearly half of patients receiving VEGFR TKIs developed hypertension.[2,7–9] Experience outside of clinical trials indicates that hypertension is observed in an even greater proportion, over 90% of patients, when ambulatory and home blood pressure (BP) monitoring is used.[10] In addition to high rates of prevalent cardiovascular disease (CVD) with heart failure (HF) in ≈6% to 9% and coronary disease in ≈12% to 20% of patients initiating VEGFR TKI therapy, rates of left ventricular dysfunction have been noted to be 10% to 28%,[11] and the cumulative incidence of major adverse cardiovascular events (MACEs) at 1 and 2 years is estimated to be 6.9% and 10%, respectively.[12,13] In a recent study of 2257 individuals receiving targeted therapy, the MACE incidence per 100 person-years was 6.65 (95% CI, 5.74–7.56), and CV death incidence was 5.55 (95% CI, 4.72–6.37).[5] The risks of MACE, including cardiomyopathy, HF, coronary disease, myocardial infarction, and stroke, are all worsened by hypertension.

However, the appropriate systolic BP (SBP) target for patients with hypertension treated with VEGFR TKIs is unknown. In the SPRINT trial (Systolic Blood Pressure Intervention Trial), which excluded patients with active cancer, lowering SBP to a target of <120 mm Hg (versus <140 mm Hg) reduced the composite risk of acute coronary syndrome, stroke, HF, or cardiovascular death by 25%.[14] These results informed the American Heart Association/American College of Cardiology recommendations for an SBP target of<130 mm Hg in individuals with clinical CVD or a 10-year CVD risk ≥10%.[15] Yet, the applicability of these guidelines to patients receiving VEGFR TKIs is unknown, as such patients are systematically excluded from cardiovascular clinical trials. Moreover, VEGFR TKI–induced hypertension is often difficult to control,[16,17] and the safety and tolerability of achieving an intensive SBP target of <120 mm Hg in patients with advanced cancer on active therapy have not been established. Consequently, less aggressive BP management is often observed in oncology practice due to concerns regarding treatment tolerance, polypharmacy, and competing clinical priorities.[18–22] To address this critical evidence gap, we conducted a phase II RCT to evaluate the feasibility of achieving and maintaining an intensive SBP target (<120 mm Hg) and to assess the safety and tolerability of this approach at the individual participant level in patients with renal cell or thyroid cancers initiating VEGFR TKI therapy. A site-based cluster randomization design was used within the ECOG-ACRIN Cancer Research Group to mitigate against the potential risk of contamination across treatment arms. Establishing the ability to safely reach this physiological target was a necessary first step toward designing a larger, definitive trial to test the efficacy of intensive SBP control in this unique population.

METHODS

Data Availability

Data may be made available upon reasonable request to ECOG-ACRIN. A request must be submitted to ECOG-ACRIN (https://ecog-acrin.org/about-ecog-acrin-cancer-research-group/organizational-overview/locations-and-contacts/).

Study Design

This phase II, open-label, site-based cluster randomized trial was conducted through the ECOG-ACRIN Cancer Research Group. Participating sites included academic medical centers and National Community Oncology Research Program (NCORP) community sites in the United States. To avoid potential between-group contamination of having both intervention (SBP <120 mm Hg, intensive control) and nonintervention (SBP <140 mm Hg, usual care) arms within a single institution, the unit of randomization was a priori defined as the site rather than the individual patient. Eligibility criteria for each site were based on participation in the ECOG-ACRIN Cancer Research Group or NCORP. Thus, each participating institution was randomized over time to either intensive SBP control or usual care (in pairs), which preceded the study participant enrollment and their consent. All participants at each institution were treated according to the randomization. All sites obtained central and institutional review board approval for the study, and each participant provided written informed consent.

The trial design of this pilot study, with the primary objective of defining the feasibility, safety, and tolerability of SBP control, called for a 6-month intervention with home BP monitoring and in-person study visits, beginning within 4 weeks of VEGFR TKI initiation. The trial was registered at ClinicalTrials.gov (NCT04467021), and the Study Protocol is included in the Supplemental Appendix.

Study Participants

Eligible patients were adults (aged ≥18 years) with an ECOG performance status of 0 to 2 and RCC or thyroid cancer initiating treatment with VEGFR TKIs, including sunitinib, sorafenib, pazopanib, cabozantinib, lenvatinib, vandetanib, or axitinib. Concurrent or prior immunotherapy was permissible. Patients who had a prior diagnosis of hypertension, were on antihypertensive medications, or those with an SBP ≥130 mm Hg on ≥2 occasions at any clinic visit in the 12 weeks before or during their initial 4 weeks of treatment with a VEGFR TKI were eligible. Key exclusion criteria included end-stage renal failure on dialysis, repeated hyperkalemia with a potassium level >5.5 mEq/L in the last 6 months, kidney transplant, an eGFR <30 mL/min per 1.73 m^2 on evaluation within 45 days before registration, cardiovascular events in the prior 3 months, and uncontrolled BP defined by SBP >160 mm Hg on ≥3 antihypertensives before TKI initiation. A full list of eligibility criteria is provided in the Study Protocol in the Supplemental Material.

Study Procedures

Upon trial registration, patients were assigned to a study arm (intensive control or usual care) according to the site randomization assignment. The baseline visit occurred within 4 weeks of VEGFR TKI initiation. At the baseline visit, training for home BP measurement and transmissions was performed. Patients were familiarized with the central BP advisory core, and questions regarding home BP monitoring were addressed. BP was measured according to a standardized protocol using the Omron 7250 or equivalent model (Study Protocol in the Supplemental Material), and questionnaires were administered. After the baseline visit, in-person clinic visits were scheduled for patients at 1, 2, 3, and 6 months, with video-assisted telehealth visits allowable during the COVID-19 pandemic. All visits included BP measurements, questionnaires, adverse event assessment, and serum chemistries. At these visits, home BP monitors, smartphones, and BP/medication logs were reviewed by the study team.

Home BP measurements were required of all patients, regardless of study arm. All patients were provided with a validated Omron 7250 oscillometric monitor[23] and detailed instructions on proper assessment of BP (Study Protocol in the Supplemental Material). Patients were instructed to take home BP readings 1 day every 2 weeks, checking their BP 4× on 1 day midweek, between 7 and 10 AM and between 7 and 10 PM, recording 2 measurements during each sitting. At least once every 2 weeks, patients were asked to submit the recorded BPs via a Bluetooth or Web-based portal to the Omron cloud; these were automatically received by the ECOG-ACRIN Statistical Center's system for Web-based data capture (EASEE patient-reported outcome [PRO]) and viewed by the site provider and the central BP advisory core. EASEE PRO was customized to integrate with the Omron cloud to capture scheduled BPs and send standardized emails and text messages with reminders and feedback regarding BP targets. All patients were provided with a list of standard recommendations for lifestyle.[24]

Central BP Advisory Core

A central BP advisory core, staffed by a nephrologist (R.R.T.) and pharmacist (PharmD; C.D.), was established as a resource to all treating physicians and coordinators, and provided additional oversight if alert BP thresholds of SBP >180 mm Hg, SBP <90 mm Hg, or diastolic BP >110 mm Hg were recorded at any time in any study patient. As detailed below, the central BP advisory core provided additional BP medication titration guidance to the site providers managing patients in the intensive SBP control arm. For providers at usual care sites, the central BP advisory core could be consulted if needed.

Intensive SBP Control

For all participants at a site (cluster) randomized to intensive SBP control, an SBP >120 mm Hg at home monitoring after the baseline visit resulted in initiation of the medication titration protocol, as described below. Typically, recommended BP medication titrations were made at 2-week intervals to allow adequate time for efficacy. If the SBP was >150 mm Hg, 2 sequential steps (ie, 2 medications) could be combined, and fixed-combination drug products could be used to decrease pill burden. A suggested algorithm for medication (class and dosage) was provided to all site providers (Study Protocol in the Supplemental Material), and medication changes were recommended by the treating physician at each site with advice from the central BP advisory core. During the titration phase (times when new drugs were added, or existing drugs were titrated), it was recommended that patients check their home BP on 3 consecutive days, with 2 measurements between 7 and 10 AM and 2 measurements between 7 and 10 PM using the instructions in the Study Protocol in the Supplemental Material. Patients were asked to upload BP recordings obtained during the titration phase 3× every 2 weeks. When patients were stable (SBP <120 mm Hg on average that week), 2 sets of readings on 1 day every 2 weeks were recorded and uploaded. The site-based research team was responsible for all patient and clinical provider communications.

Usual Care

For all participants at a site (cluster) randomized to usual care, patients were instructed to take home BP readings 1 day every 2 weeks in the morning and evening of the same day. Additional BP measurements were not required in this group unless instructed by their treating physician. There was no titration phase. It was suggested to participants that their home BP goal should be <140 mm Hg systolic and, on average, be 135 to 139 mm Hg systolic, but this was not mandated. Patients received antihypertensive management at their clinic sites consistent with standard care in the medical practice where they received care. The same BP medications used in the intensive SBP control arm to treat hypertension could be used in the usual care arm according to the discretion of the clinical site provider. At the baseline visit, if the SBP was >160 mm Hg, it was advised according to study protocol that antihypertensive therapy be initiated for safety purposes. If the SBP was <160 mm Hg at the baseline visit, the addition or titration of BP medications was at the discretion of the treating provider. The central BP advisory core was also a resource for the sites, as needed.

Questionnaires

At baseline, 1, 2, 3, and 6 months, the following questionnaires were administered to the patient by the site research staff (Supplemental Methods).

Medication Adherence

The validated PROMIS Medication Adherence scale, version 1.0 (2/19/2020), included 12 items that assessed patient comprehension of the purpose for taking medication, perceptions of the importance of taking this medication, self-reported adherence, and reasons for nonadherence.[25]

Patient Satisfaction With BP Care and Medications

Two items from the SPRINT trial were administered to obtain patient ratings of satisfaction with BP care and BP medications.[26,27] Eleven NCI PRO-Common Terminology Criteria for Adverse Events (CTCAE) items provided a standardized assessment of 5 patient-rated symptomatic adverse events, including swelling (frequency, severity, and interference), palpitations (frequency and severity), dyspnea (severity and interference), dizziness (severity and interference), and fatigue (severity and interference).[28]

CTCAE Targeted Reporting

All toxicity grades described throughout this protocol and all reportable adverse events were graded using the NCI CTCAE, version 5.0. In addition, site provider ratings for the following targeted adverse events were obtained to ensure the ascertainment of common potential side effects and cardiovascular toxicities associated with antihypertensive treatment: acute kidney injury, chest pain-cardiac, dehydration, dizziness, dyspnea, fatigue, generalized edema, headache, HF, hyperkalemia, hypertension, hypokalemia, hypotension, left ventricular systolic dysfunction, presyncope, and stroke.

Primary and Secondary End Points

The primary end point was the difference in SBP in the intensive arm versus the usual care arm over the 6-month study period based on automated study visit (clinic) BP measurements. Our primary end point, reported at the individual level according to arm, was based on our primary aim of determining the feasibility of our approach to intensive SBP control in this active cancer population and not on the widespread implementation of our approach into clinical practice. Secondary end points, reported at the individual participant level according to arm, included the difference in SBP over the 6-month study period based on all automated home and clinic BP measurements; study retention rates; frequency of automated home SBP measurements >180 or <90 mm Hg or diastolic BP >110 mm Hg; the frequency of hypotension; and differences in symptomatic adverse events, patient-reported medication compliance, patient-rated symptomatic adverse events, patients' acceptance, and satisfaction with BP medications according to the Treatment Satisfaction Questionnaire for Medication questionnaire (version 1.4). Medication compliance via the PROMIS Medication Adherence scale and PRO-CTCAE item scores were also summarized.

Statistical Analysis

Because the primary objective was to define the feasibility, safety, and tolerability of an intensive versus usual care approach to SBP management, our sample size was based on a detectable difference in SBP. Sample size calculations were originally performed assuming independent participants (intraclass correlation coefficient [ICC], 0). Here, a sample size of 30 patients per arm provided the study with ≈80% power to test the hypothesis that patients randomized to the intensive SBP control arm would have a 15-mm Hg lower average SBP than those randomized to the usual care arm over the 6-month period. This assumed a 2-sided alpha of 0.05, an average SBP of 130 mm Hg at baseline, a SD of 18 mm Hg, and a within-person correlation (between baseline and 6-month follow-up) of 0.40. We further assumed that study attrition would be minimal (<5%, ie, resulting analysis sample size=29 per arm). Additional details regarding the ICC calculation and power analysis evaluating the sensitivity to ICC[29] are noted in the Supplemental Methods.

The primary end point, the difference in clinic SBP over the 6-month study period between intensive SBP control versus usual care, was compared using the difference-in-difference method with 95% CIs generated using the bootstrap method[30] (with 5 study sites as the resampling units [per each randomization arm] to account for within site correlations and 5000 resamples). Specifically, for any cross-sectional comparisons, we used bootstrapping with the cluster/site as the primary resampling unit to account for within-cluster correlation. For analysis evaluating the differences in SBP over time, we needed to account for both within-patient and within-cluster correlation in 6-month changes. We first calculated individual change scores (which effectively collapses 2 longitudinal measurements into a single value) and then, based on the calculated changes, performed bootstrapping using the cluster/site as the basic resampling unit. Patients with missing measurements at a visit were excluded from that specific timepoint analysis. As a secondary analysis, the SBP trajectories based on all automated home and clinic BP measurements were determined using a similar approach.

Results for all other secondary outcomes were summarized using descriptive statistics and graphical displays, without hypothesis testing, given the limited sample size and exploratory nature of the analyses. Across all primary and secondary end points, data were complete, with <5% missing data, except for the PROMIS Medication Adherence scale questionnaire in the usual care arm. No interim analyses comparing the study arms were performed, and patients' study arm assignments were blinded to the principal investigator until study analyses' completion.

RESULTS

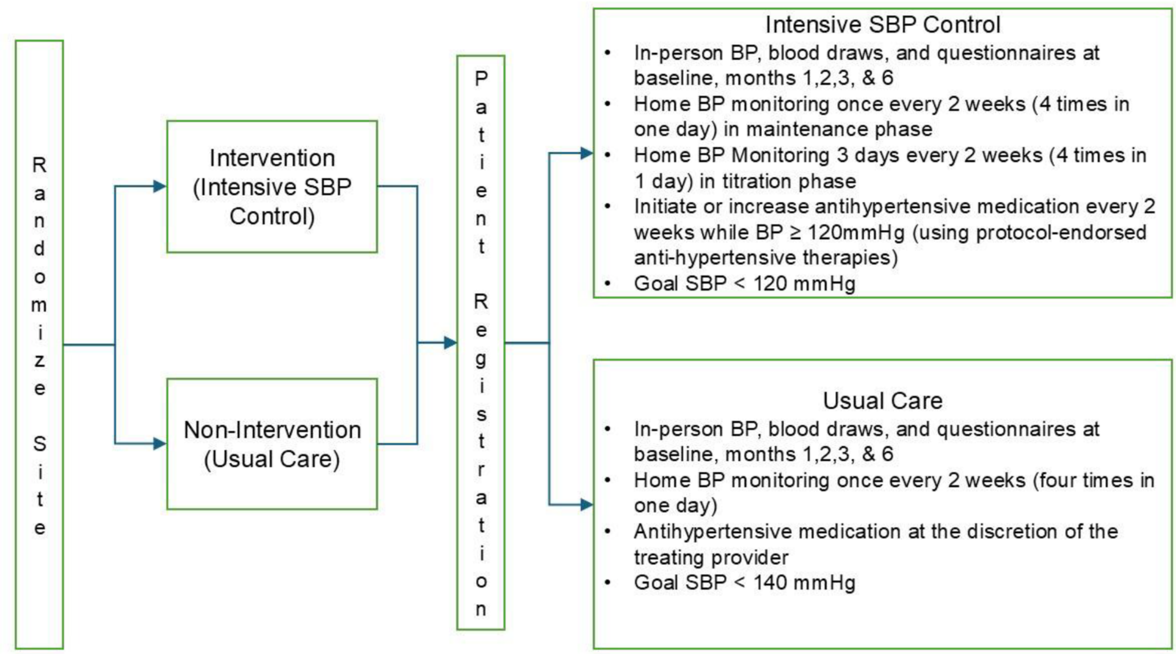
From September 2020 to December 2023, 10 sites were randomized (Figure 1; Figure S1). Of these, 7 sites were academic institutions, and 3 were NCORP community-based sites. Overall, 61 patients were recruited, with 30 from 5 sites randomized to intensive SBP control and 31 from 5 sites randomized to usual care (Table S1). Of these, 58 patients (95.1%) had RCC, and 3 patients (4.9%) had thyroid cancer. Retention rates were high and were similar across both arms, with rates of 100% at 1 and 2 months in the intensive SBP control arm and 96.8% in the usual care arm. The proportions decreased slightly to 89.7% at 6 months in the intensive SBP control arm and to 93.3% in the usual care arm; 54 patients (91.5%) completed the 6-month study visit (Figure 2).

The baseline demographic and clinical characteristics of the patients were overall well-balanced across both groups (Table 1). Most patients had prevalent hypertension (96.7% in the intensive SBP control and 93.5% in the usual care arm) before VEGFR TKI initiation. Diabetes was common with 30.0% in the intensive and 22.6% in the usual care arms. Overall, 8.2% had prevalent HF, with 3 in the intensive SBP control and 2 in the usual care arms. Coronary disease was present in 20% in the intensive SBP control and 12.9% in the usual care arms.

Differences in SBP: Primary and Secondary End Points

The mean and median clinic SBPs in each arm over time are noted in Table 2. At baseline, the mean SBP was lower in the intensive SBP control arm compared with usual care although the median was similar. At each subsequent timepoint, the mean and median clinic SBPs were lower in the intensive SBP control arm (Table 2; Figure 3A). We observed a significantly lower clinic SBP measured at the 1- and 3- month visits in the intensive SBP control arm compared with the usual care arm, with mean differences between the 2 arms of −12.2 (95% CI, −18.1 to −7.0) mm Hg at 1 month and −7.6 (95% CI, −15.3 to −0.4) mm Hg at 3 months. At 2 months, this difference was −6.7 (95% CI, −16.4 to 4.4) mm Hg and, at 6 months, −6.9 (95% CI −19.3 to 6.0) mm Hg (Figure 3B). In comparing the change from baseline, the greatest differences were observed at 1 month (−8.0 [95% CI −15.8 to 1.4] mm Hg). The observed ICC of SBP across participants over time (baseline to 6 months) was 0.068 (95% CI, 0.004–0.366).

No primary end point BP measurements exceeded the safety threshold of SBP <90 mm Hg or SBP >180 mm Hg. Over the 6-month study period, depending on the timepoint, 3.4% to 10.7% of patients in the usual care arm reached a threshold of SBP >160 mm Hg (Table 2) based on clinic BP measurements. The mean number of BP medications at each timepoint is displayed in Figure 3A. As expected, the mean number of BP medications was consistently greater in the intensive SBP control arm compared with usual care (Figures 3A and 3B). The distribution of BP medications according to drug class and over time is shown in Table S2. The overall number of planned medication dose changes (dose reduced, increased, or held) that were recorded at each visit was greater in the intensive SBP control arm compared with usual care, with 24 to 39 medication dose changes and 1 to 7 medication dose changes, respectively, recorded at each study visit across all participants over 6 months.



Sites:

Intensive SBP Control	Usual Care
University of Pennsylvania	Montefiore NCORP
Aurora Health Care NCORP	Washington University
University of Texas at Southwestern	University of Pittsburgh Medical Center
Fox Chase Cancer Center	Virginia Commonwealth University NCORP
City of Hope	University of Massachusetts

Figure 1. Study design.
This figure details the overall study design for this site-based cluster randomized phase 2 clinical trial. BP indicates blood pressure; NCORP, National Community Oncology Research Program; and SBP, systolic blood pressure.

Home BP values mirrored those of the clinic BPs, with consistently lower SBPs over time in the intensive SBP control arm compared with usual care (Table S3; Figure 3C) and a greater prevalence of SBPs exceeding a threshold of 160 mm Hg in the usual care arm.

Adverse Events

Rates of grade 3 to 5 CTCAEs are summarized in Table 3. There was 1 grade 4 hypertension event in the usual care arm that was determined not to be related to the study intervention. No other grade 4 to 5 intervention-related adverse events were reported across any participants, including potential side effects and cardiovascular adverse events associated with antihypertensive medications. In the usual care arm, grade 3 CTCAEs for acute kidney injury (n=4), hypotension (n=3), and dyspnea (n=2) were numerically greater compared with the intensive SBP control arm (n=1, 2, and 0, respectively). Two deaths were noted, one in each arm, with one in the intensive SBP control arm being secondary to disease progression and the other in the usual care arm due to an unspecified reason. Both, however, were not related to the study.

Patient-Reported Symptomatic Adverse Events

The distribution of PRO-CTCAEs over time is displayed in Figure S2. Qualitatively, dyspnea was more prevalent

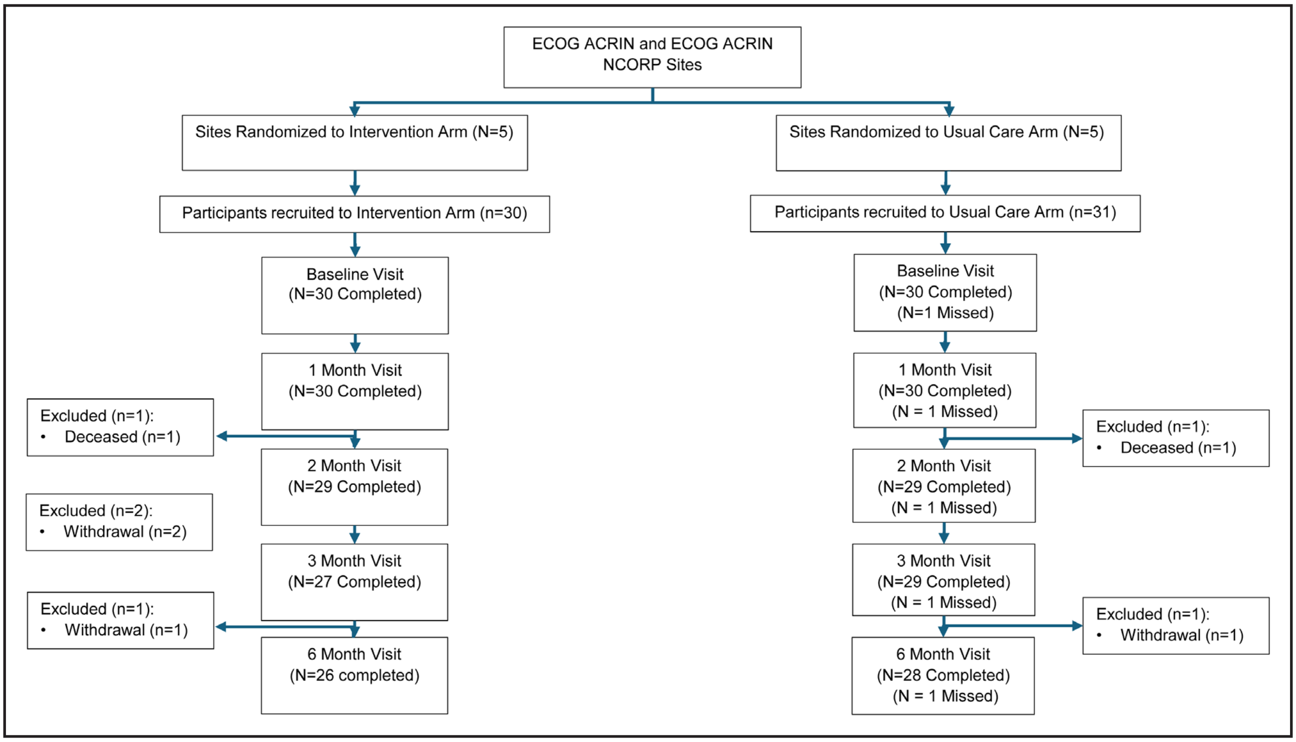


Figure 2. Consort diagram.
This figure details the consort diagram. NCORP indicates National Community Oncology Research Program.

in the intensive SBP control arm (frequency of any dyspnea: 50% at 1 month, 40.7% at 2 months, 44% at 3 months, and 57.7% at 6 months) compared with usual care (40.7%, 30.8%, 37%, and 37%, respectively). However, across all timepoints, swelling was consistently less prevalent in the intensive SBP control arm (any swelling 17.9% at 1 month, 22.2% at 2 months, 8% at 3 months, and 23.1% at 6 months) compared with usual care (29.6%, 34.6%, 25.9%, and 22.2%, respectively). Otherwise, there were no consistent differences in other PRO-CTCAEs.

Patient Satisfaction and Adherence

Overall satisfaction was high with the BP medications used in the study, with a very low proportion of patients overall expressing dissatisfaction and a greater proportion noting dissatisfaction in the usual care arm compared with the intensive SBP control arm (Figure S3). A low proportion overall noted side effects with BP medications, and this distribution was similar between the 2 arms (Figure S4). Medication adherence was similar between groups with a PROMIS Patient Medication adherence score that ranged from 41.4 to 43.1 in the intensive SBP control arm and 41.8 to 43.4 in the usual care arm although there was a greater proportion of missing data in the usual care arm, on the order of 3.8% to 12.9% across the various timepoints. In contrast, in the intensive SBP control arm, the proportion of missingness was 0% to 3.8%.

DISCUSSION

This cluster randomized trial is the first to demonstrate the feasibility, safety, and tolerability of an intensive SBP control approach in an active, advanced cancer population with hypertension receiving VEGFR-targeted cancer therapy. Using cluster randomization at the site level across 10 academic and NCORP community–based sites, we were able to initiate and titrate antihypertensive therapy for the intensive management of BP in patients with hypertension receiving VEGFR TKIs and achieve a significant lowering of BP compared with usual care. At 1 month, SBP was on average 12.2 mm Hg lower in the intensive SBP control arm, and this reduction was consistent over the 6-month study period. As hypertension associated with VEGFR TKI therapy is often severe and resistant to multiple antihypertensive medications,[16,17] the feasibility of intensive SBP control in our study was not a predetermined outcome. The ability to achieve our primary end point underlines the effectiveness and innovation of our approach, including the use of home BP monitoring, the use of a platform for centralized BP monitoring and management, the implementation of a central BP advisory core with a detailed hypertension management algorithm, and collaboration across the multi-disciplinary team at both academic and community oncology sites.

VEGFR TKIs are the standard of care singly or in combination with immunotherapy in RCC and thyroid cancer, and have resulted in substantial improvements

Table 1. Baseline Patient Clinical Characteristics

	Intensive	Usual care	Total
Sites, N	5	5	10
No. of participants/cluster (site), median (range); mean	4 (2–14); 6	5 (2–13); 6.2	4.5 (2–14); 6.1
Participant, N	30	31	61
Age, y; mean (SD)	65.7 (5.77)	64.5 (8.33)	65.1 (7.15)
BMI, mean (SD)	32.3 (8.11)	29.5 (5.17)	30.9 (6.86)
Male	21 (70.0%)	26 (83.9%)	47 (77.0%)
Smoking status			
Current/former	12 (40.0%)	13 (41.9%)	25 (41.0%)
Race			
Black	1 (3.3%)	3 (9.7%)	4 (6.6%)
Not reported or unknown	2 (6.7%)	2 (6.5%)	4 (6.6%)
White	27 (90.0%)	25 (80.6%)	52 (85.2%)
History of diabetes	9 (30.0%)	7 (22.6%)	16 (26.2%)
History of hypertension	29 (96.7%)	29 (93.5%)	58 (95.1%)
History of arrhythmia	6 (20.0%)	6 (19.4%)	12 (19.7%)
History of heart failure*	3 (10.0%)	2 (6.5%)	5 (8.2%)
History of coronary artery disease	6 (20.0%)	4 (12.9%)	10 (16.4%)
Type of cancer			
Renal cell cancer	30 (100.0%)	28 (90.3%)	58 (95.1%)
Thyroid cancer	0 (0.0%)	3 (9.7%)	3 (4.9%)
Cancer stage			
Stage 1			
T1N0	1 (3.3%)	0 (0.0%)	1 (1.6%)
T1NX	0 (0.0%)	1 (3.2%)	1 (1.6%)
T1Unknown N	1 (3.3%)	0 (0.0%)	1 (1.6%)
Stage 2			
T2N0	1 (3.3%)	1 (3.2%)	2 (3.3%)
T2NX	2 (6.7%)	0 (0.0%)	2 (3.3%)
Stage 3			
T3N0	3 (10.0%)	5 (16.1%)	8 (13.1%)
T3N1	1 (3.3%)	2 (6.5%)	3 (4.9%)
T3NX	9 (30.0%)	6 (19.4%)	15 (24.6%)
Stage 4			
T4 any N M0	1 (3.3%)	0 (0.0%)	1 (1.6%)
T4 any N M1	1 (3.3%)	4 (12.9%)	5 (8.2%)
Any T any N M1	7 (23.3%)	10 (32.3%)	17 (27.9%)
Unknown stage	3 (10.0%)	2 (6.5%)	5 (8.2%)
TKI types			
Axitinib	14 (46.7%)	8 (25.8%)	22 (36.1%)
Cabozantinib	11 (36.7%)	12 (38.7%)	23 (37.7%)
Lenvatinib	5 (16.7%)	8 (25.8%)	13 (21.3%)
Pazopanib	0 (0.0%)	1 (3.2%)	1 (1.6%)
Tivozanib	0 (0.0%)	2 (6.5%)	2 (3.3%)
Immune checkpoint inhibitor therapy			
Nivolumab	8 (26.7)	6 (19.4)	14 (23.0)
Pembrolizumab	13 (43.3)	11 (35.5)	24 (39.3)
Any antihypertensive medication	29 (96.7%)	28 (93.3%)	57 (93.4%)

(*Continued*)

Table 1. Continued

	Intensive	Usual care	Total
Antihypertensive medication			
ACE inhibitor	5 (17.2%)	8 (28.6%)	13 (22.8%)
ARB	17 (58.6%)	5 (17.9%)	22 (38.6%)
β-blocker	15 (51.7%)	13 (46.4%)	28 (49.1%)
Calcium channel blocker	16 (55.2%)	16 (57.1%)	32 (56.1%)
Diuretic	8 (27.6%)	7 (25%)	15 (26.3%)
Nitrates	1 (3.4%)	0 (0.0%)	1 (1.8%)
α-Blocker	0 (0.0%)	0 (0.0%)	0 (0.0%)
Other	0 (0.0%)	2 (7.1%)	2 (3.5%)

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BMI, body mass index; and TKI, tyrosine kinase inhibitor.

*In the intensive arm, 1 patient had heart failure with reduced ejection fraction, and 2 patients had heart failure with preserved ejection fraction. In the usual care arm, both patients had heart failure with reduced ejection fraction.

in oncological mortality, with overall survival rates on the order of 46 to 56 and 26 to 36 months in RCC and thyroid cancer, respectively. However, patients receiving these therapies are at significantly increased risk of MACE, attributed at least in part to significant hypertension that is a common cardiovascular toxicity of these therapies. For example, the likelihood of developing a symptomatic decrease in left ventricular ejection fraction is 2-fold higher in those with a median SBP ≥140 mm Hg on VEGFR TKIs (hazard ratio, 2.1 [95% CI, 1.3–3.4]).[31] However, the impact of BP control, and how this may mitigate CVD risk in cancer patients, has remained untested, in part secondary to concerns over BP lowering in active cancer. Our trial has overcome this important knowledge gap through our demonstration of the feasibility, safety, and tolerability of intensive SBP control.

In SPRINT, intensive SBP control to a target goal of <120 mm Hg resulted in a 38% reduction in HF, a 43% reduction in cardiovascular death, and a 27% reduction in all-cause death.[14] In the ACCORD trial (Action to Control Cardiovascular Risk in Diabetes), intensive SBP control was associated with a 41% lower incidence of stroke than in patients with higher SBP targets.[32] Although the aim of our trial was not to detect differences in cardiovascular outcomes, and the primary objective was to determine the feasibility, safety, and tolerability of an intensive SBP control approach in an active, advanced cancer population, to contextualize our pilot trial findings, the differences we achieved in the intensive SBP control and usual care arms, if corroborated in subsequent studies, are potentially meaningful. In the general population, meta-regression analyses of RCTs with a minimum of 1000 patient-years of follow-up have demonstrated that each 10-mm Hg reduction in SBP significantly reduced the risk of major cardiovascular events (overall relative risk, 0.80 [95% CI, 0.77–0.83]), coronary heart disease (0.83 [95% CI, 0.78–0.88]), stroke (0.73 [95% CI, 0.68–0.77]), HF (0.72 [95% CI, 0.67–0.78]), and all-cause mortality (0.87 [95% CI, 0.84–0.91]).[33] Moreover,

Table 2. Clinic Blood Pressures Across All Timepoints According to Study Arm

	Baseline		1 mo		2 mo		3 mo		6 mo	
	Intensive	Usual care	Intensive	Usual care	Intensive	Usual care	Intensive	Usual care	Intensive	Usual care
N, %	30 (100%)	30 (96.8%)	30 (100%)	30 (96.8%)	29 (96.7%)	29 (93.5%)	27 (90%)	29 (93.5%)	26 (86.7%)	28 (90.3%)
SBP, mm Hg										
Mean (SD)	133.6 (16.8)	137.6 (17.4)	126.2 (11.9)	138.3 (15.2)	125.3 (11.6)	131.9 (15.9)	122.0 (12.6)	129.6 (15.9)	124.3 (14.0)	131.2 (17.5)
Median	133.0	133.5	124.5	139.0	123.0	130.0	122.0	129.0	123.5	128.0
Min	101	97	108	112	103	99	102	101	100	105
Max	179	179	148	173	159	162	149	164	158	172
Threshold										
<140	19 (63.3%)	19 (63.3%)	24 (80.0%)	16 (53.3%)	26 (89.7%)	21 (72.4%)	24 (88.9%)	23 (79.3%)	22 (84.6%)	21 (75.0%)
<120	7 (23.3%)	2 (6.7%)	9 (30.0%)	4 (13.3%)	9 (31%)	6 (20.7%)	12 (44.4%)	9 (31%)	10 (38.5%)	7 (25.0%)
<90	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
>180	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
>160	2 (6.7%)	4 (13.3%)	0 (0)	3 (10%)	0 (0)	1 (3.4%)	0 (0)	1 (3.4%)	0 (0)	3 (10.7%)

SBP indicates systolic blood pressure.

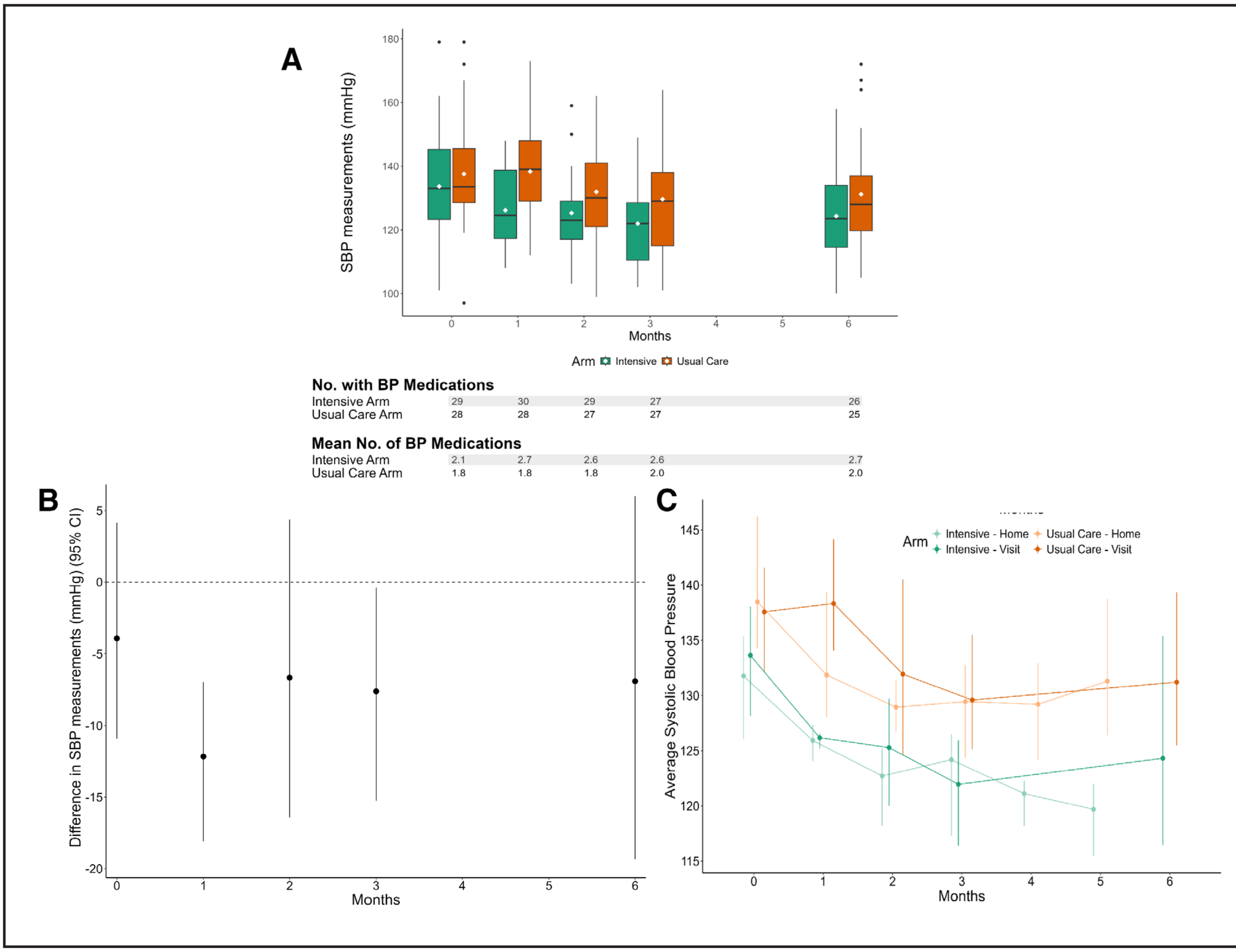


Figure 3. Blood pressure (BP) differences according to treatment arm.

A, Distribution of systolic BPs (SBPs) according to treatment arm and timepoint. This plot shows the distribution of SBP measurements for each visit by arm using boxplots. The white diamonds represent the average systolic measurements for each timepoint, while the black line shows the median SBP. **B**, The difference in average SBP at each timepoint between arms. This plot shows the difference (intensive SBP control arm—usual care arm) in average SBP measurements at each timepoint with the 95% CI from the bootstrapped results shown in Table 2. If the difference is lower than 0, this indicates that the average SBP at the timepoint is lower in the intensive SBP control arm compared with the usual care arm. **C**, Comparing average SBPs at home and clinic according to treatment arm and timepoint. This plot shows the average SBP per arm by visit and home and clinic measurements over time. The 95% CIs are also displayed and computed through a bootstrapped generalized intercept-only model with clustering by the 5 sites in each randomization arm within the arm for 5000 iterations.

in recent randomized clinical trials such as SPRINT and BPROAD (Blood Pressure Control Target in Diabetes), the separation of Kaplan-Meier curves for the primary cardiovascular outcome between the intensive treatment group and the standard treatment group was apparent after 1 year of the BP intervention,[14,34] suggesting potential clinical relevance to a population receiving anticancer therapy with a median survival that largely exceeds this. A recent secondary analysis suggested that, on average, 9.1 (95% CI, 4.0–20.6) months were needed to prevent 1 MACE per 500 patients with the intensive BP treatment (absolute risk reduction, 0.002).[35]

Of note, SBP trends recorded by patients during home self-monitoring mirrored those of the clinic SBP recordings and demonstrate the durability of our intervention between clinic visits. Moreover, there was no worsening in grade 3 or higher adverse events with intensive SBP control, and some were even numerically greater with usual care, demonstrating the safety of our intervention. In terms of patient-reported symptomatic adverse events, these were similar across both groups with a lesser prevalence of swelling among those with intensive SBP control. Patients demonstrated strong compliance with study procedures with little dropout. Moreover, patients demonstrated high satisfaction with this approach, and those in the intensive SBP control arm noted better BP medication compliance. Part of the reticence among oncologists, cardiologists, primary care providers, and patients for SBP lowering in clinical practice is the concern for causing harm and adverse side effects from intensive SBP management and more medications beyond what is necessary.[36] However, our findings suggest that patients

Table 3. Grade 3 or Worse Targeted Adverse Events and Deaths According to Treatment Arm at Any Timepoint

CTCAE	Intensive	Usual care
Hypotension	2	3
Acute kidney injury	1	4
Chest pain	0	0
Dehydration	2	0
Dizziness	1	0
Dyspnea	0	2
Fatigue	4	2
Edema	0	0
Headache	0	0
Hyperkalemia	0	0
Hypokalemia	1	1
Presyncope	0	0
Stroke	0	0
Heart failure	0	0
Left ventricular systolic dysfunction	1	0
Death	1	1

The following targeted adverse events were ascertained: acute kidney injury, chest pain-cardiac, dehydration, dizziness, dyspnea, fatigue, generalized edema, headache, heart failure, hyperkalemia, hypokalemia, hypotension, left ventricular systolic dysfunction, presyncope, and stroke. CTCAE indicates Common Terminology Criteria for Adverse Events.

tolerated intensive SBP control well and had numerically fewer concerns than expected in an advanced cancer population. These findings provide a scientific premise for future trials with larger sample sizes, and our approach provides a potential strategy that can be adapted.

As with any study, there are limitations. The foremost is the sample size. This resulted in some differences in our baseline covariates. Moreover, with our limited sample size and power, we acknowledge that certain effects did not reach statistical significance, but the observed trends offer more valuable insights and can help guide further investigation in future large studies. While our initial sample size calculations did not account for ICC, our power calculations that evaluated the sensitivity to the ICC indicate the study still maintained sufficient power for meaningful interpretation of our outcomes. We further emphasize effect estimates and corresponding CIs. Given the pilot study nature, this provides a more informative and nuanced view of the observed associations than reporting *P* values. Moreover, given the small sample size per cluster and limited number of clusters, we did not use longitudinal data analytic approaches that are often preferred to capture within-subject trends but rather evaluated the individual change scores (which effectively collapses 2 longitudinal measurements into a single value) and then, based on the calculated changes, performed bootstrapping using the cluster/site as the basic resampling unit. Our study duration was relatively short and was not designed to evaluate the impact of intensive SBP control on CVD outcomes. The site-based cluster randomization design, used to avoid contamination in the usual care approach, resulted in slowed patient enrollment. We also cannot exclude the possibility that site-specific practice patterns based on personal preferences may have impacted some of our results. This study was also open-label, but, to our knowledge, no recent intensive SBP control studies have been blinded. We also fully acknowledge that this was not an implementation RCT but was designed as a pilot study to generate the necessary and sufficient data to inform a larger, definitive RCT. However, we also note multiple strengths. This is the first-ever study in an active cancer population with cancer treatment–associated hypertension that reached its primary end point of a significant difference in SBP with intensive SBP control compared with usual care. Moreover, our study was both innovative and rigorous through its use of a central BP advisory core, home BP monitoring, and integration of a platform that was able to receive home BPs, allow for biweekly patient notifications and reminders to increase compliance of BP measurements, and issue alerts for patients and study staff of any BP alarm thresholds. We also had geographic and institutional diversity, recruiting patients across the country from both university- and community-based health systems to emphasize relevance with trends in national cancer care delivery. Our findings lay the necessary foundation to test the implementation of an Intensive SBP approach in large-scale intervention trials evaluating cardiovascular outcomes in active cancer populations, which are our proposed next steps.

In conclusion, this first-ever study of intensive SBP control in an active cancer population with hypertension receiving VEGFR TKIs demonstrates the feasibility, safety, tolerability, and acceptability of intensive SBP control among cancer patients with elevated cardiovascular risk secondary to their cancer treatment.

PERSPECTIVES

This cluster randomized trial is the first to demonstrate the feasibility, safety, and tolerability of intensive SBP control in an active, advanced cancer population with hypertension receiving targeted cancer therapy regimens. Using cluster randomization at the site level across 10 academic and NCORP community-based sites, we were able to initiate and titrate antihypertensive therapy for the intensive management of BP in patients with hypertension receiving VEGFR TKIs and achieve a significant lowering of BP compared with usual care.

ARTICLE INFORMATION

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Supplemental Material

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